

Akshay Jaiswal

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Summary

Software Engineer with 4+ years of experience in scalable systems, cloud infrastructure, and DevOps, currently pursuing M.S. in Computer Science. Experienced in building scalable systems and applying deep learning techniques for generative modeling and computer vision. Skilled in automating ML pipelines and deploying production-ready AI solutions.

Professional Experience

Software Engineer | Tata Consultancy Services

Nov 2020 – Jan 2025

Mumbai, India

- Led the migration of legacy on-prem banking systems to AWS, improving scalability, achieving 99.9% uptime, and reducing infrastructure costs by 25% while ensuring compliance with security and regulatory standards.
- Engineered and optimized cloud infrastructure and CI/CD pipelines, reducing deployment time by 40% and increasing release frequency from weekly to daily builds.
- Drove incident response and resolution, handling 50+ production incidents/month and reducing MTTR by 30% through root cause analysis and cross-team coordination.

Intern | Tata Consultancy Services

Jan 2020 – Jul 2020

Nagpur, India

- Designed and developed RESTful APIs for an employee management system, and improving data retrieval efficiency by 35%.
- Built backend services using Java, Spring Framework, Maven, and Tomcat to enable CRUD operations and streamline workflows, reducing manual data handling effort by 40%.

Education

Master of Science in Computer Science

Jan 2025 – Present

California State University, Long Beach

GPA: 3.667

Bachelor of Engineering in Computer Science

Aug 2016 – Jun 2020

Gyan Ganga Institute of Technology & Sciences

CGPA: 3.49

Projects

ChestX-AI Assistant & Pneumonia Detection | Agentic AI, Computer Vision, LLMs

Jan 2026 – Present

- Developed Computer Vision and Transfer Learning models (DenseNet-121, ResNet-18) for multi-label classification of 5 thoracic diseases using 191K+ chest X-ray images from 45K+ patients, achieving 0.75+ mean AUROC.
- Designed a multi-model, multi-agent AI pipeline integrating 3 LLMs (OpenAI, Gemini, Groq) with Computer Vision models to transform diagnostic probabilities into structured radiology reports, reducing manual interpretation effort and improving report consistency.
- Built an evidence-grounded Agentic AI architecture leveraging Prompt Engineering, Explainable AI (Grad-CAM), and structured JSON reasoning to generate 4-section clinical reports while mitigating hallucinations and supporting clinical decision-making.

QuizLet | CSU AI Hackathon (IBM SkillsBuild)

Mar 2026

- Developed an AI-powered learning assistant within 24 hours during the CSU AI Hackathon, leveraging Generative AI, Prompt Engineering, IBM Watson, IBM Orchestrate, and Ollama to automate quiz generation and personalized learning workflows.
- Collaborated with a cross-functional team to build and demo a functional prototype for the Education & Workforce Intelligence track, applying Machine Learning, Intelligent Automation, Streamlit, Python, and Responsible AI principles under IBM mentorship.

Deep Learning with PyTorch: GANs – hands-on model implementations

Sept 2025 – Jan 2026

- Designed and trained deep learning models, generating 10K+ synthetic images and improving output quality (20% FID reduction) through iterative experimentation and evaluation.
- Developed GAN-based pipelines with stability techniques and hyperparameter tuning, reducing training instability by 30% and improving convergence speed by 25%, while producing visually consistent outputs across applications such as car design and handwritten image generation.

Technical Skills

Programming Languages: Python, Java, JavaScript, C++, C#, PHP

Machine Learning: PyTorch, CNNs, GANs, ResNet, NumPy, Matplotlib, CUDA

Cloud & DevOps: AWS (EC2, S3, IAM, Lambda, RDS, DynamoDB, API Gateway, CloudFormation, CloudWatch), Docker, Terraform, Ansible, CI/CD (GitHub Actions, Jenkins, AWS CodePipeline)

Databases: PostgreSQL, DynamoDB, Oracle, SQL

Tools: Git, GitHub, Jira, ServiceNow, Bitbucket, TeamDynamix

Training and Certifications

Cisco Networking Academy: Cybersecurity, CCNA, IoT

Jan 2018

TCS-ION Digital Learning Hub: Core Java (34 hours)

Jan 2018

Ozone-soft Solutions: LAMP (Linux, Apache, MySQL, PHP) training (1 week)

Jun 2019